Connected Rates of Change

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Help me understand

Connected rates of change

What are connected rates of change?

- In situations involving more than two variables you can use the **chain rule** to connect multiple rates of change into a single equation

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

ALL TIMES, FIND THE RATE OF CHANGE OF THE SIDE LENGTH, S, IN cm PER SECOND AT THE MOMENT WHEN S = 5 cm.

$\frac{dV}{dt} = 3 \text{ cm}^3/\text{sec}$

WE KNOW THIS

WE WANT TO KNOW THIS

BY THE CHAIN RULE $\frac{dV}{dt} = \frac{dV}{ds} \times \frac{ds}{dt}$

USE THE CHAIN RULE TO CONNECT WHAT YOU KNOW AND WHAT YOU WANT TO KNOW

FOR A CUBE, $V = s^3$

SO $\frac{dV}{ds} = 3s^2$

USE DIFFERENTIATION TO GET THE 'MISSING' DERIVATIVE FROM THE CHAIN RULE EQUATION

SO $\frac{ds}{dt} = \frac{\frac{dV}{dt}}{\frac{dV}{ds}} = \frac{3}{3s^2} = \frac{1}{s^2}$

REARRANGE

WHEN $s = 5$,

$\frac{ds}{dt} = \frac{1}{5^2} = \frac{1}{25}$ cm/sec

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- Equations involving derivatives (i.e. rates of changes) are known as **differential equations**
  - These can be solved using methods of integration (see Differential Equations)
  - However setting up the equation from the information given can involve the chain rule and connected rates of change

e.g. A RIGHT-ANGLED TRIANGLE HAS A BASE OF **3p** cm AND A HEIGHT OF **2p** cm, WHERE **p > 0** IS A PARAMETER THAT IS INCREASING OVER TIME. THE RATE OF INCREASE OF **p** IS SUCH THAT THE RATE OF INCREASE OF THE THE AREA OF THE TRIANGLE, **A**, IS PROPORTIONAL TO THE SQUARE ROOT OF **p**. WRITE DOWN A DIFFERENTIAL EQUATION FOR THE RATE OF CHANGE OF **p**.

$\frac{dA}{dt} \propto \sqrt{p}$ SO $\frac{dA}{dt} = k\sqrt{p}$

A IS INCREASING SO $k > 0$

WHERE k IS A POSTITVE CONSTANT

WE KNOW THIS

BY THE CHAIN RULE, $\frac{dA}{dt} = \frac{dA}{dp} \times \frac{dp}{dt}$

USE THE CHAIN RULE TO CONNECT WHAT YOU KNOW AND WHAT YOU WANT TO KNOW

WE WANT TO KNOW THIS

THE AREA OF THE TRIANGLE IS

$A = \frac{1}{2} \times 3p \times 2p = 3p^2$

USE DIFFERENTIATION TO GET THE 'MISSING' DERIVATIVE FROM THE CHAIN RULE EQUATION

SO $\frac{dA}{dp} = 6p$

THEN $\frac{dA}{dt} = \frac{dA}{dp} \times \frac{dp}{dt}$

SUBSTITUTE AND REARRANGE

$k\sqrt{p} = 6p\left(\frac{dp}{dt}\right)$

AND $\frac{dp}{dt} = \frac{k\sqrt{p}}{6p}$

$\frac{dp}{dt} = \frac{k}{6\sqrt{p}}$ (WHERE k IS A POSITIVE CONSTANT)

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- Note from the examples above that you will often need to differentiate a formula to get one of the rates of change

Examiner Tips and Tricks

- These problems can involve a lot of letters – be sure to keep track of what they all refer to.
- Be especially sure that you are clear about which letters are **variables** and which are **constants** – these behave very differently when differentiation is involved!

Worked Example

In a manufacturing process, plastic spheres are produced in such a way that the volume, V, of a sphere increases at a constant rate of $10\ \text{cm}^3$ per second. Find the rate of change of the surface area, A, of a sphere at the moment when the surface area is equal to $32\pi\ \text{cm}^2$.

$\frac{dV}{dt} = 10\ \text{cm}^3/\text{sec}$

WE KNOW THIS

WE WANT TO KNOW THIS

BY THE CHAIN RULE $\frac{dV}{dt} = \frac{dV}{dA} \times \frac{dA}{dt}$

USE THE CHAIN RULE TO CONNECT WHAT YOU KNOW AND WHAT YOU WANT TO KNOW

$V = \frac{4}{3}\pi r^3$ (WHERE r IS THE RADIUS)

WRITE V IN TERMS OF A

$A = 4\pi r^2$

$A^{\frac{3}{2}} = (4\pi r^2)^{\frac{3}{2}} = 8\pi^{\frac{3}{2}} r^3 = 8\pi\sqrt{\pi}\, r^3$

$\frac{8\pi\sqrt{\pi}\, r^3}{6\sqrt{\pi}} = \frac{4}{3}\pi r^3$ SO $V = \frac{1}{6\sqrt{\pi}} A^{\frac{3}{2}}$

AND $\frac{dV}{dA} = \frac{3}{2} \times \frac{1}{6\sqrt{\pi}} A^{\frac{3}{2}-1} = \frac{1}{4\sqrt{\pi}} A^{\frac{1}{2}}$

DIFFERENTIATE TO FIND THE 'MISSING' DERIVATIVE

SO $\frac{dV}{dt} = \frac{dV}{dA} \times \frac{dA}{dt}$

SUBSTITUTE AND REARRANGE

$10 = \frac{1}{4\sqrt{\pi}} A^{\frac{1}{2}} \left(\frac{dA}{dt}\right)$

AND $\frac{dA}{dt} = \frac{40\sqrt{\pi}}{A^{\frac{1}{2}}} = 40\sqrt{\frac{\pi}{A}}$

WHEN $A = 32\pi$,

$\frac{dA}{dt} = 40\sqrt{\frac{\pi}{32\pi}} = \frac{40}{\sqrt{32}} = 5\sqrt{2}\ \text{cm}^2/\text{sec}$

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